

Modelling Particle Production in Analog Expanding Universes with High-Performance Computing

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Introduction - Analog Universes

① Inflation created particles

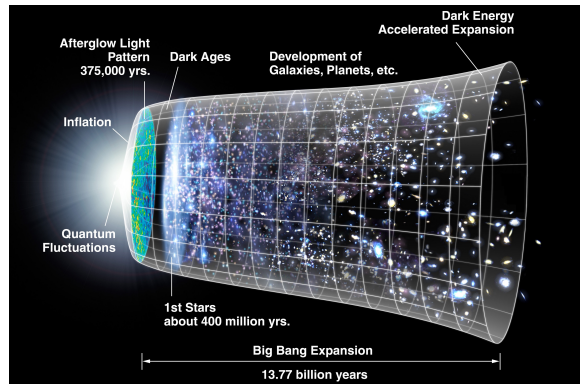


Figure: History of the universe highlighting period of rapid expansion (Source)

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- 2 Cannot directly observe these particles

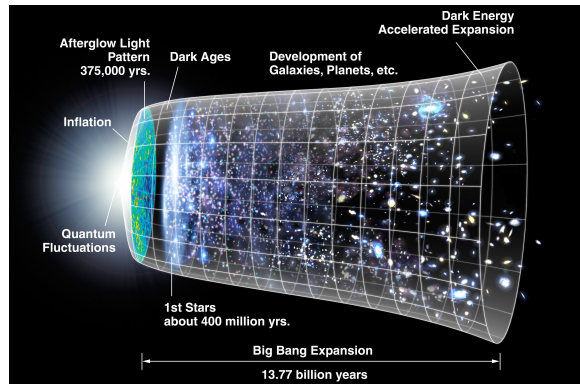


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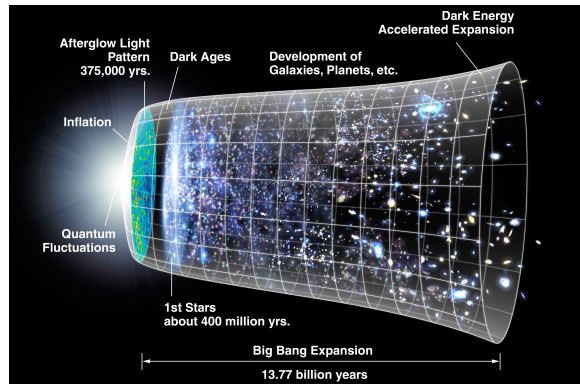


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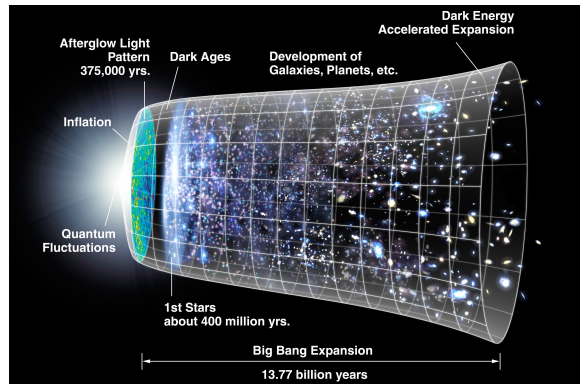


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- 5 Lab based system = “analog universe”

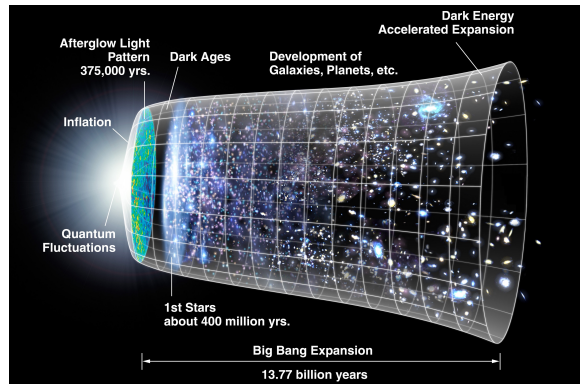


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- Many investigations into analog universes



Figure: Subsets of increasingly specific literature

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- Fewer into computational simulation
- Even fewer using high-performance computing



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